Measuring an Equilibrium Constant Results

Part 1. Measuring initial concentrations

Table I: Experimental concentrations and light absorbance Co & HCl

Test tube number	0.100M Co^{+2}	12.0M HCl (Cl^- ions)	Total volume	Absorbance
1	2.0ml	8.0ml	10.0 ml	0.950
2	3.0ml	7.0ml	10.0 ml	0.830
3	4.0ml	6.0ml	10.0 ml	0.600
4	5.0ml	5.0 ml	10.0 ml	0.350

Use the data from the above table to determine the concentration of the reactants after dilution. $M_iV_i = M_fV_f$

Use the absorbance along with the molar absorptivity (ϵ 600 1/M) to determine equilibrium concentrations of $CoCl_4^{-2}$ Abs = €* C

Table II: Measuring Equilibrium concentrations of $CoCl_4^{-2}$

Test tube number	Co^{+2} conc (M) from dilution (Initial)	Cl^- conc(M) from dilution (Initial)	Equil. $CoCl_4^{2-}$ conc(M) from absorbance
1	0.0200	9.6	0.00158
2	0.0300	8.4	0.00138
3	0.0400	7.2	0.001
4	0.0500	6.0	0.000583

Part 2. Calculating K

For each of the four reactions: use the data from Table 2 to set up an ICE table for the $CoCl_4^{-2}$ reaction and then solve for K

Lab 5 Part I

$M_f = 0.1M$

$V_i \rightarrow$ varies

$V_f = 10\,mL$

Column 1

$M_1V_1 = M_fV_f$

$\frac{M_1V_1}{V_f} = M_f = \frac{(0.1)(2)}{10} = \frac{0.1}{5} = 0.02M$

$\frac{(0.1)(3)}{10} = \frac{0.3}{10} = 0.03M$

$\frac{(0.1)(4)}{10} = 0.04M$

$\frac{(0.1)(5)}{10} = 0.05M$

Column 2

M_1 12 M

V_1 8 mL → changes

V_f 10 mL

$M_1V_1 = M_fV_f$

$\frac{(12)(8)}{10} = 9.6$

$\frac{(12)(6)}{10} = 7.2$

$\frac{(12)(7)}{10} = 8.4$

$\frac{(12)(5)}{10} = 6.0$

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S	M	T	W	T	F	S
26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23/30	24	25	26	27	28	29

$Abs = \varepsilon C$

$C = \frac{Abs}{\varepsilon} \rightarrow$ 0.950 here

$C = \frac{Abs}{600}$

- $\frac{0.950}{600} = 0.0015\,M$
- $\frac{0.830}{600} = 0.00138\,M$
- $\frac{0.600}{600} = 0.001\,M$
- $\frac{0.350}{600} = 0.000583\,M$

Part II

$$Co^{2+} + 4Cl^- \rightleftharpoons CoCl_4^{2-}$$

Tube 1

from prev table $x = 0.00158M$

	Co^{2+}	$4Cl^-$	$CoCl_4^{2-}$
i	0.0200	9.60	0
Δ	$-x$	$-4x$	$+x$
⇌	$0.0200 - x$	$9.60 - 4x$	x
	$0.0200 - 0.00158$	$9.60 - 4(0.00158)$	0.00158
	0.0184	9.59	

$$K = \frac{[CoCl_4^{2-}]}{[Co^{2+}][Cl^-]^4} = \frac{0.00158}{(.0184)(9.59)^4}$$

$$\frac{0.00158}{(0.0184)(8458)} = \frac{0.00158}{155.6} = 10.2 \times 10^{-6}$$

$$10.2 \times 10^{-6}$$

Trial 2

$x = 0.00138$

	Co^{2+}	$4Cl^-$ $\rightleftharpoons$	$CoCl_4^{2-}$
I	0.03	8.40	0
Δ	$-x$	$-4x$	$+x$
E	0.0286	8.39	0.00138

$0.0286 - 0.00138$ $\quad\quad$ $8.39 - 4(0.00138)$

0.0286 $\quad\quad$ 8.39

$$K = \frac{0.00138}{(0.0286)(8.39)^4} = \frac{0.00138}{141.7} = 9.74 \times 10^{-6}$$

Tube 3 $x = 0.00100$

	Co^{2+}	$4Cl^-$	$CoCl_4^{2-}$
i	0.04	7.2	0
Δ	−0.001	−4(0.001)	+0.001
$\rightleftharpoons$	0.00390	7.2	0.001

$$K = \frac{0.001}{(0.0390)(7.20)^4} = \frac{0.00100}{1048} = \underline{9.54 \times 10^{-6}}$$

$x = 0.000583$

Tube 4

	Co^{2+}	$4Cl^-$	$CoCl_4^{2-}$
i	0.05	6	0
Δ	−0.000583	$-4x$	$+x$
$\rightleftharpoons$	0.0494	6	0.000583

$$K = \frac{0.000583}{(0.0494)(6)^4} = \frac{0.000583}{64} = \underline{9.11 \times 10^{-6}}$$

All values are close to each other.